*Core areas:* Computer Vision and Machine Perception; Safety, Reliability, and Robustness; Reinforcement Learning and Data-driven Control.
*Impact areas:* AI for Robotics and Autonomous Systems and Physical AI; AI for Space Research and Exploration.

Learned state estimators can be more accurate than classical filters, but unlike a Kalman filter they rarely report when they are wrong. I have met this problem in every project since my Master's. In my doctorate I want to give learned estimators on legged robots an uncertainty that stays calibrated when conditions change, can be checked without ground truth, and can be used by a controller.

**How I arrived at this question.** I came to machine learning from classical navigation. As an avionics student I built a GPS-aided inertial navigation system on an extended Kalman filter, where every estimate arrives with a covariance. At the University of Tehran I reproduced IONet, a deep inertial odometry model, and then turned to estimating full attitude from raw IMU data. My first models silently assumed a fixed interval between samples. With my advisor, Prof. M. H. Sabour, I redesigned the network to take the sampling rate as an explicit input, much as a numerical integrator needs its time step. The resulting models, published in *Measurement* (2023), transferred across devices and outperformed classical filters on several datasets. On others they lost to the strongest neural baseline, and nothing in their output indicated which case applied.

From 2023 to 2025 I worked part-time as a navigation engineer at Fasta Robotics, where the problem moved from one sensor to a complete robot. I migrated the robots' software from ROS 1 to ROS 2, integrated the navigation stack, and deployed ORB-SLAM3-based visual–inertial SLAM with LiDAR on warehouse robots and a Unitree Go2 quadruped, using RealSense D435i and ZED 2i cameras. In real warehouses, poor lighting, repetitive shelving and forklifts crossing the camera's view degraded the visual input. ORB-SLAM3 reports no uncertainty that navigation can use, so a degraded pose looked the same as a good one until tracking was lost.

My most recent paper, TG-DCM-PINN (Turkish Robotics Conference, 2026), returned to the single IMU with a narrower idea. Physics-informed training usually applies its constraints everywhere, but an accelerometer measures gravity only when the body is not accelerating. A gravity constraint applied during fast motion therefore teaches the model something false. My model applies each physical constraint only where the measurements support it, and weights the gravity term by a confidence computed from acceleration, jerk and rotation rate. On 137 sequences from five public datasets this gave a small but statistically reliable reduction in mean tilt error over the strongest neural baseline. On a harder benchmark, one segment in twenty still exceeded 44°. The confidence that tells the model during training when its physics can be trusted is discarded at deployment. I did this work independently, outside a university and alongside full-time employment, and released the code and evaluation protocol.

**What is missing.** Classical consistency checks, such as innovation tests, detect an inconsistent filter, but they assume correctly modelled Gaussian errors, which learned components violate. Recent conformal methods give Kalman filters calibrated confidence regions, but they calibrate on trajectories with ground-truth states and assume exchangeable data (Weisman et al., 2026). Rønning et al. (2026) showed that disagreement between two estimators can expose overconfidence without ground truth. Their test is one-sided, assumes Gaussian errors, and loses power when the estimators' errors are correlated. In Prof. Leutenegger's visual–inertial SLAM, learned depth uncertainty is calibrated with a single gain fitted on validation data, and the authors name epistemic uncertainty and closing the loop with control as next steps (Jung et al., 2025). The open question, as I see it, is how a robot can maintain calibrated, two-sided uncertainty for its learned estimators during deployment, when ground truth is absent and conditions shift.

**Proposed research.** I want to turn the redundancy of a legged robot into a calibration signal. Leg kinematics, several IMUs, cameras and LiDAR observe the same motion with partly independent errors. The method I would develop is a conformal calibration in which nonconformity scores are computed against a reference fused from the other sensors, rather than against ground truth. I expect its guarantee to degrade by an amount set by the reference's own error; establishing that bound is part of the work. The scores would be conditioned on the motion regime, using signals such as the acceleration, jerk and rotation-rate confidence from my own estimator. Changes in the score distribution over time would provide the estimate of shift that robust conformal prediction needs.

The GrandTour dataset from Prof. Hutter's group makes each step testable. It records an ANYmal at alpine, forest, urban and indoor sites with seven IMUs, joint encoders, LiDARs, D435i and ZED 2i cameras, and millimetre-level ground truth. That ground truth would be used only to validate the calibration, never to fit it. The first step measures how far a single global gain is from calibrated across these sites. The second builds and tests the cross-sensor

calibration. Its main risk is correlated failure, as when every sensor degrades in darkness or on slippery ground. Characterising when the reference fails is itself a result, because it marks the limit of any redundancy-based guarantee. The third step passes the calibrated bounds to a controller, so that the robot slows down or seeks better observations when it is uncertain.

**Why these mentors and the ETH AI Center.** The project needs two fields in equal measure, which the Center's two-mentor model supports. Prof. Hutter's group has worked on knowing when a sensor cannot be trusted. X-ICP detects directions in which a LiDAR scan cannot constrain the pose, and lists degeneracy-aware fusion as future work (Tuna et al., 2024). My aim is the counterpart for learned components, which lack any comparable geometric test, and GrandTour is the natural testbed. Prof. Leutenegger's learned-uncertainty SLAM is the estimator in which a fixed gain could be replaced by deployment-time calibration. Prof. Lindemann's work provides the statistical side and the link to control. His conformal bounds on state-estimation error enable safe perception-based control, but they are calibrated against reference states (Yang et al., 2023). His recent work recovers coverage under shift when the size of the shift can be estimated (Rahaman et al., 2026). Removing the reference-state requirement, and estimating the shift from the robot's own sensors, is where my project would add to these lines rather than repeat them. I would hope to be co-supervised by Prof. Hutter or Prof. Leutenegger together with Prof. Lindemann.

**Strengths for the impact areas, and interdisciplinary work.** My background spans avionics, space engineering and machine learning, and I have deployed localization on real robots. Space makes this problem sharper. Hutter's group has operated teams of ANYmal robots in planetary-analogue terrain (Arm et al., 2023). On a real mission there is no ground-truth system and no operator to re-tune an estimator. At the MIT IAIFI AI+Physics program in 2024 I used sequential neural posterior estimation, a simulation-based inference method that is close to the statistical thinking this project needs. In industry I have worked in mixed teams of engineers and domain experts, and I led a team of six engineers at AI-Ark.

**Technology transfer and community.** A company like Fasta cannot collect ground truth in every warehouse or replace a working estimator. A calibration layer that needs neither is the form in which this research could reach industry, and it is why I am interested in the Center's entrepreneurial track. For the community, I would release the calibration benchmark built on GrandTour so that other fellows can test their estimators against it, as I have done with the code and evaluation protocols of my papers. I review for IEEE RA-L and have supervised five undergraduate theses, and I would continue mentoring students at the Center.

**Career goals.** I want an academic research career in reliable autonomy for field and space robots. Working alone, I can improve a single estimator. The questions above need multi-sensor legged robots, data from many environments, and supervision in both estimation and statistics, and this fellowship provides all three.

**Cited works.**


- Asgharpoor Golroudbari, A., & Sabour, M. H. (2023). Generalizable end-to-end deep learning frameworks for real-time attitude estimation using 6DoF inertial measurement units. *Measurement*, 217, 113105.
- Asgharpoor Golroudbari, A. (2026). Target-gated direction-cosine physics for six-axis inertial attitude estimation. Turkish Robotics Conference.
- Weisman, O., Shlezinger, N., & Laufer-Goldshtein, B. (2026). Conformalized Kalman filters for state estimation with trustworthy confidence regions. arXiv:2609.27506.
- Rønning, O., Saqib, U., & Waşowski, A. (2026). You should be properly scoring your odometry. arXiv:2609.25900.
- Jung, J., Boche, S., Barbas Laina, S., & Leutenegger, S. (2025). Uncertainty-aware visual-inertial SLAM with volumetric occupancy mapping. *IEEE ICRA*, 14550–14556.
- Tuna, T., Nubert, J., Nava, Y., Khattak, S., & Hutter, M. (2024). X-ICP: Localizability-aware LiDAR registration for robust localization in extreme environments. *IEEE Transactions on Robotics*, 40.
- Tuna, T., Frey, J., et al. (2026). GrandTour: A legged robotics dataset in the wild for multi-modal perception and state estimation. arXiv:2602.18164.
- Yang, S., Pappas, G. J., Mangharam, R., & Lindemann, L. (2023). Safe perception-based control under stochastic sensor uncertainty using conformal prediction. *IEEE CDC*, 6072–6078.
- Rahaman, K., Deshmukh, J. V., Hota, A. R., & Lindemann, L. (2026). When environments shift: Safe planning with generative priors and robust conformal prediction. *L4DC*.
- Arm, P., et al. (2023). Scientific exploration of challenging planetary analog environments with a team of legged robots. *Science Robotics*.